

DUST-BATHING IN THE DOMESTIC FOWL — DIURNAL RHYTHM AND DUST DEPRIVATION

KLAUS VESTERGAARD

Royal Veterinary and Agricultural University, Department of Forensic and State Veterinary Medicine, 13, Bülowsvej, DK-1870 Copenhagen V (Denmark)

(Accepted for publication 19 March 1981)

ABSTRACT

Vestergaard, K., 1982. Dust-bathing in the domestic fowl — diurnal rhythm and dust deprivation. *Appl. Anim. Ethol.*, 8: 487–495.

The diurnal rhythm of dust-bathing and the effect of dust deprivation on this behaviour were studied in a flock of 14 White Leghorn egg-layers. Increasing deprivation time lowered latency and raised duration of dust-bathing. Latency was longer when measured in the morning (09.15 h) than in the afternoon (14.15 h), but time of day did not significantly affect duration. Individual differences were found for duration but not for latency. When the birds had continual access to litter, their dust-bathing behaviour showed a marked peak around mid-day from 12.00 h to 13.00 h, and on average birds performed one 27-min bout of dust-bathing every second day.

INTRODUCTION

Dust-bathing is found within many groups of birds and descriptions of dust-bathing components are available for many species, including the domestic fowl (Simmons, 1964). However, the function and regulatory mechanism of dust-bathing are still open to question.

Three functions have been suggested. (1) Removal of surplus lipids from the plumage, as was demonstrated in Bobwhite quail (*Colinus virginianus*) by Borchelt and Duncan (1974). The lipids are secreted by the uropygial gland and dabbed onto the feathers during preening. (2) Improvement of feather structure has been demonstrated in the Japanese quail (*Coturnix coturnix japonica*) by Healy and Thomas (1973). Dust-bathing improved barb alignment of the feathers, made the down more fluffy and dry, and reduced the amount of dandruff in the feathers. (3) Removal of ectoparasites has been suggested as the primary function of dust-bathing (e.g. Simmons, 1964) but never experimentally proved.

Effects of deprivation from dust have been studied in the Bobwhite quail (Borchelt et al., 1973; Borchelt, 1975) and in the Japanese quail (Benson and Schein, 1965). Generally, deprivation resulted in an increased tendency to per-

form dust-bathing after deprivation, as some elements increased in number with increasing deprivation time.

Domestic hens in cages and on sloped wire floors are constantly without access to dust and the consequences of this have been much debated (Hughes, 1973; Black and Hughes, 1974; Martin, 1975; and others). However, little knowledge of the effect of dust deprivation is available, both concerning the consequences of the lost function and the psychic effect of the lack of releasing stimulus. The aim of the present study was to provide more basic knowledge of the long-term temporal regulation of dust-bathing by examining the effect of increasing deprivation time and of giving access to dust (litter) at different times of the day.

MATERIALS AND METHODS

Animals and housing

Fifteen White Leghorn hens, 18 weeks old, were purchased at a commercial dealer. They had been beak-trimmed at one day old and then raised in a large flock in a deep-litter house. In the laboratory, the hens were marked with wing marks and marking rings and until 43 weeks of age they were housed in a deep-litter pen 3×3.2 m. Thereafter, they were marked by speed-marker colours on tail, head and back, and transferred to the experimental room (Fig. 1).

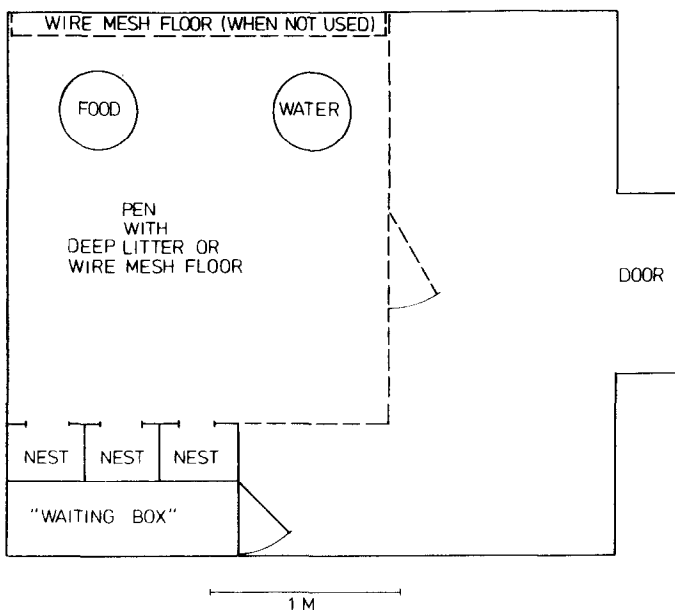


Fig. 1. Plan of the experimental room. The wire-mesh floor was normally placed against the wall but covered the litter during deprivation periods. While the floors were exchanged, the hens were placed in the "waiting box" outside the pen.

The floor of the pen consisted of deep-litter (straw), but the rest of the room was of polished concrete (terrazzo floor). Along one of the walls in the pen there was a wire mesh which, by lowering, covered the floor 20 cm above the deep litter. There were three nests with straw. The hens were fed *ad libitum* with a layer ration from a feeding trough, and water was available from an automatic waterer. The room was provided with automatic ventilation, which maintained the temperature at $22 \pm 2^\circ\text{C}$. Artificial light, from a 60 watt light bulb and four 40 watt neon tubes (Osram 32), was on from 06.15 h to 20.15 h. The illumination at floor level was 250 lux. There was no additional source of illumination from windows, etc. One hen became ill and was removed before the tests began.

Deprivation

Before the experiments, the hens were trained to move to the "waiting box" (see Fig. 1) where they stayed for 5 min. They also became accustomed to floor changes from litter to wire or vice versa at their return to the pen. Throughout the experiment the "waiting box" procedure was followed every week-day morning, including days of deprivation of dust. Before each deprivation period there were at least two days with continual access to deep-litter. During deprivation, the floor was covered with a sheet of black plastic to prevent visual cues from the litter, and above this the wire-mesh floor was placed. Inspection revealed that the hens never performed dust-bathing when the wire floor was in use. Deprivation started at 09.15 h, after the visit to the "waiting box". "Morning tests" were made at 09.15 h after deprivation for 24, 48 and 96 h, and "afternoon tests" at 14.15 h were made after 5, 53 and 101 h, respectively. Each of the six deprivation levels were repeated five times with the whole flock. The order of the treatments was presented at random. After half of the experiments were made, there was a period of nine weeks without tests and with continual access to litter. During this period, the diurnal rhythm of the birds was observed. The deprivation experiments were continued after one week re-training of the "waiting box" procedure. Throughout the study, the hens were maintained and tested in the flock, which meant that before any test all the 14 birds had been exposed to the same deprivation level.

Because nesting behaviour might interfere with dust-bathing, it was noted during the first hour after the end of deprivation if any hen stayed in one of the nests. Birds in the nests were allowed to remain there during removal or insertion of the wire floor.

When deprivation was to be terminated, the hens were guided to the "waiting box" and after 5 min they were allowed to return to the pen, where meanwhile the wire-mesh floor and the plastic sheet had been removed to give free access to the litter.

Latency for the occurrence of the first "vertical wing-shake" (Kruijt, 1964) was recorded individually. Duration of dust-bathing was then timed until the first "body/wing-shake" (Kruijt, 1964). Records were taken only for hens

which started dust-bathing within the first hour of observation. During the observations, one person observed the animals and dictated to another who made the notes and checked the time. Besides these tests, a written record of the general behaviour was made.

Diurnal rhythm

The diurnal rhythm of dust-bathing was examined by continual observation in the light periods over ten days after half of the deprivation experiment had been carried out. Six weeks prior to, and during, the ten days of observation, the hens had unrestricted access to the deep litter and in this period also the visits to the "waiting box" were suspended. The time at the onset, as well as the duration, of dust-bathing were recorded individually, as during the deprivation experiments.

Statistical analysis

Differences between treatments in the proportions of hens which initiated dust-bathing during the first hour after the end of deprivation were statistically evaluated by the Wilcoxon matched-pairs signed-ranks test (Siegel, 1956). Each hen was given one score if it initiated dust-bathing. Thus, for each type of treatment, a hen could obtain up to 5 scores, any of the six treatments being repeated five times. On this basis, the treatments were compared, two by two, by using each hen as its own control.

The effects of deprivation-time and time of day (morning versus afternoon) on latency and duration of dust-bathing were evaluated by analysis of covariance. The covariable was deprivation time, and independent factors were time of day and hen.

The influence of nesting on the latency and duration of dust-bathing was evaluated by application of the Sign test (Siegel, 1956). Each hen was used as its own control by comparing the latency and duration at tests when the hen performed nesting (and dust-bathing) with that of the following test at the same deprivation level when it performed dust-bathing but no nesting.

Correlation between the mean duration of the dust-bathing bout of each hen, obtained during the diurnal rhythm observations, with that obtained during the deprivation tests was evaluated by calculation of the Spearman's rank-correlation coefficient (Freund, 1974) and testing for significance. The means from the deprivation tests were those adjusted in the analysis of covariance for the effect of deprivation-time and time of day.

RESULTS

General observations during deprivation tests

After deprivation, nearly all the hens immediately began to peck and scratch

as during feeding on the floor, but more eagerly. After a while, the birds lowered their body while still pecking and scratching and then became more stationary. At this stage, they very often raised the upper tail covers and thereafter most of the birds soon squatted in the litter and began to perform the dust-bathing. The following elements were observed: (1) bill-raking; (2) scratching; (3) vertical wing-shaking; (4) head-rubbing; (5) lying on the side.

Before squatting in the dust, interruptions were common. The bird then rose, walked around pecking and scratching, and finally settled in another place. This might happen several times. Some birds pecked, scratched, and lowered their body, but failed to proceed with other dust-bathing activities.

When the birds had performed the first vertical wing-shake, interruptions were rare and they normally continued dust-bathing for several minutes. Even during this phase, they might be forced to shift position because of aggression from other birds, but in these cases they immediately found another place to settle and continue their dust-bathing.

At the end of the dust-bathing sequence, the hen rose to its feet and after a few seconds it performed feather-raising closely followed by vigorous shaking of its body (body/wing-shake).

The birds tended to dust-bathe close together, and dust-bathing birds were targets for pecking and scratching by the other birds.

Deprivation

The mean proportions of birds starting dust-bathing within the first hour with access to litter after deprivation are given in Fig. 2 (left). At the morning tests there was a gradual increase with deprivation; 48 h of deprivation gave significantly higher proportions than 24 h, and 96 h significantly higher proportions than both 24 and 48 h of deprivation ($P < 0.01$). At the afternoon

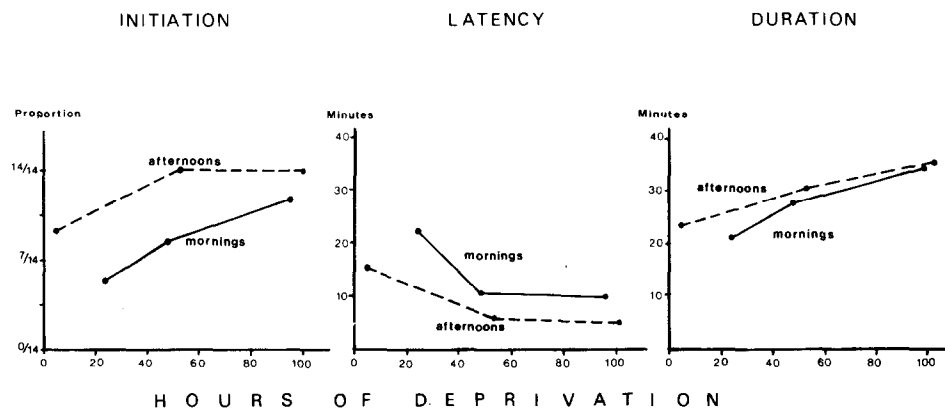


Fig. 2. Effects of deprivation time and time-of-day on (1) proportions of hens initiating dust-bathing (left), (2) latency of dust-bathing (centre), and (3) duration of dust-bathing (right).

TABLE I

Analysis of covariance of the effects of deprivation time, hen, and time of day on latency and duration of dust-bathing

Effect	DF	F values	
		Latency	Duration
Deprivation time	1	32.5***	76.2***
Hen	13	1.0	3.4***
Time of day	1	15.0***	0.1
Hen $\times$ time of day	13	1.1	1.0

*** = $P < 0.001$.

tests, an increase from 5 to 53 h of deprivation was observed ($P < 0.01$). After 53 and 101 h, the proportions were very close to the maximum possible.

Generally, the proportions of hens starting at the morning tests were smaller compared to those of the afternoon tests, but the effect of time of day could not be directly evaluated because the deprivation levels at the morning tests differed from those at the afternoon tests. However, significantly smaller number of hens initiated dust-bathing on the mornings following 24 h of deprivation than on the afternoons following 5 h of deprivation ($P < 0.01$). Also, 48 and 96 h of deprivation had a smaller effect on the morning performance than 53 and 101 h had on the afternoon performance ($P < 0.05$). There was no significant difference between the proportions found at the morning tests after 48 h of deprivation and the afternoon tests after 5 h of deprivation. Over-all dust-bathing occurred in 308 cases out of the 420 possible.

The analysis of covariance (Table I and Fig. 2 centre and right) indicated significant reduction of the latencies with increased deprivation time, and shorter latencies during afternoon tests but no significant effects due to differences among hens. Duration increased with deprivation time, but there were no time-of-day effects. Differences among hens, however, had a significant effect on duration of dust-bathing.

Twenty-five cases of nesting behaviour were recorded during the deprivation tests. Of these, 16 cases of nesting behaviour occurred without being preceded or followed by dust-bathing, whereas 9 cases of nesting behaviour preceded or followed dust-bathing. Thus, in the 25 cases where nesting occurred, dust-bathing was much less frequent than in the 395 cases where no nesting occurred (36 and 74%, respectively). In the six cases where nesting occurred before dust-bathing, the latencies were much larger than at the following comparable tests with no nesting behaviour (16–60 versus 2–5 min; $P < 0.05$), but in the three cases where nesting occurred after dust-bathing, the latencies were similar to those obtained when no nesting was performed. When nesting occurred together with dust-bathing, the duration was significantly shorter than at the following comparable tests when no nesting was seen (10–35 versus 29–44 min; $P < 0.05$).

Diurnal rhythm

Dust-bathing occurred from 08.50 h until 19.25 h and there was a marked peak in the initiation of dust-bathing between 12.00 h and 13.00 h (Fig. 3). During the observation period, all 14 hens performed dust-bathing. On average,

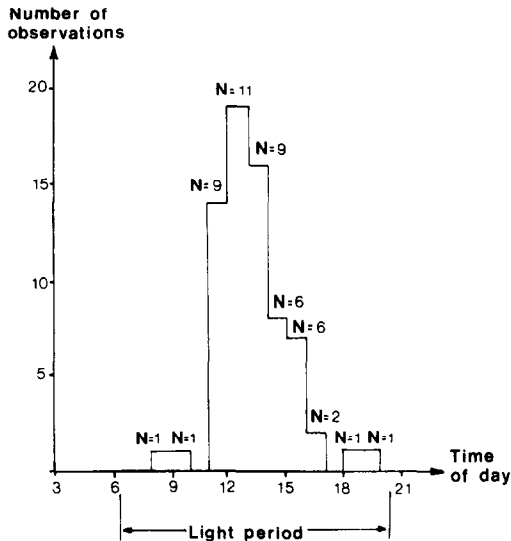


Fig. 3. Diurnal rhythm in the initiation of dust-bathing during 10 consecutive days with continual access to litter. N indicates the number of different hens which contributed to the columns.

the hens dust-bathed on five out of the ten observation days; there was, however, a large variation ranging from 2 to 9 days. The over-all mean duration of the dust-bathing bouts was 27 min, but there was a large individual variation ranging from 20 to 35 min. The mean bout-duration of each hen was positively correlated with that obtained during the deprivation tests ($r = 0.75$; $P < 0.01$).

DISCUSSION

The dust-bathing elements which were observed in this study have previously been described by Kruijt (1964) in the Red Jungle fowl. In the domestic fowl, Nice (1962) observed the same elements, except for lying on the side. The feather-raising and body/wing-shaking have been described in the Red Jungle fowl (Kruijt, 1964), and the present study indicates that these elements always terminate dust-bathing in the domestic fowl. This is also the case in the Bobwhite quail (Borchelt et al., 1973).

In the Bobwhite quail, Borchelt et al. (1973) found that two dust-bathing components (head-rub and dust-toss) increased in numbers with deprivation over five days, but latency was not significantly affected. These results were confirmed by Borchelt (1975), who found that additionally five elements al-

so increased in frequency over 1–5 days of deprivation. Similarly, Benson and Schein (1965) found that more dust-bathing was seen in Japanese quails kept on wire-mesh floors and tested in sawdust as compared to birds kept and tested in sawdust. Wennrich (1977) placed a deep-litter flock of hens in battery cages for three weeks, and at the return to litter there was an increased frequency of dust-bathing compared to the frequency prior to caging. In the cages, the dust-bathing frequency was much reduced. The present data from domestic hens are in agreement with these earlier findings, since the reduced latency and the increased duration and proportion due to deprivation can all be taken as measures of an increased tendency to perform dust-bathing.

A gradual increase of the dust-bathing tendency with the length of deprivation, as found in this study, may explain the occurrence of vacuum activities in the domestic fowl kept in battery cages (e.g. Black and Hughes, 1974) and on sloped wire floors (Vestergaard, 1981). The absence of vacuum dust-bathing in the present study may be due to the relatively short deprivation time used.

The demonstrated accumulation of the tendency to perform dust-bathing may reflect an internal motivational mechanism responsible for the regulation of this behaviour. On the other hand, a theory has been put forward (Borchelt et al., 1973) that dust-bathing happens in order to regulate the amount of lipids in the plumage, and that the lipid is the exteroceptive cue that elicits dust-bathing when the amount rises above the threshold level. In a later paper (Levine et al., 1974), however, it was shown that in the Bobwhite quail, the amount of oil in the feathers could not be the only factor responsible for dust-bathing behaviour, because dust-bathing occurred a long time after the uropygial glands were removed. Thus, the nature of the regulatory mechanism is still open to question, and the existence of a central mechanism remains a possibility.

In this study, dust-bathing showed a clear diurnal rhythm during the light hours. Known diurnal rhythms of behaviour patterns in the domestic fowl include mating (Penquite et al., 1929), drinking (Siegel and Guhl, 1956; Wood-Gush, 1959), preening, feeding and sleeping (Wood-Gush, 1959). No previous records for dust-bathing rhythm in the domestic fowl seem to exist, but a diurnal rhythm in dust-bathing similar to the present one has been observed in the Japanese quail (Statkiewicz and Schein, 1980).

The (internal?) diurnal rhythm which was demonstrated when the hens had continual access to dust seems to be the explanation for the time-of-day effect in the deprivation tests. Thus, the lower tendency for dust-bathing in the morning tests was in agreement with the very small amount of dust-bathing occurring before 11.00 h during the period with continual access to dust.

Nesting behaviour was found to influence the dust-bathing performance during deprivation tests. Dust-bathing was less frequent when nesting occurred, and latency and/or duration was altered when the hen performed both dust-bathing and nesting during the test. However, nesting behaviour during deprivation tests was rare, and is not considered to have influenced the main conclusions drawn from the deprivation experiment.

ACKNOWLEDGEMENTS

I wish to thank the following people at the Department of Forensic and State Veterinary Medicine, Royal Veterinary and Agricultural University, Copenhagen: Dr. H.C. Adler for encouraging and providing facilities for the study, Dr. P. Willeberg for help with statistical analysis, and Miss B. Jørgensen for technical assistance. The study was supported by a grant from the Danish Agricultural and Veterinary Research Council.

REFERENCES

- Benson, B.N. and Schein, M.W., 1965. Factors influencing dust bathing in *Coturnix* quail. (Motion picture). *Am. Zool. Abstr.*, 5: 196.
- Black, A.J. and Hughes, B.O., 1974. Patterns of comfort behaviour and activity in domestic fowls: A comparison between cages and pens. *Br. Vet. J.*, 130: 23–33.
- Borchelt, P.L., 1975. The organization of dust bathing components in Bobwhite quail (*Colinus virginianus*). *Behaviour*, 53: 217–237.
- Borchelt, P.L. and Duncan, L., 1974. Dust bathing and feather lipid in Bobwhite quail (*Colinus virginianus*). *Condor*, 76: 471–472.
- Borchelt, P.L., Eyer, J. and McHenry, D.S., Jr., 1973. Dust bathing in Bobwhite quail (*Colinus virginianus*) as a function of dust deprivation. *Behav. Biol.*, 8: 109–114.
- Freund, J.E., 1974. *Modern Elementary Statistics*. Prentice-Hall, London, 532 pp.
- Healy, W.M. and Thomas, J.W., 1973. Effects of dusting on plumage of Japanese quail. *Wilson Bull.*, 85: 442–448.
- Hughes, B.O., 1973. Animal welfare and the intensive housing of domestic fowls. *Vet. Rec.*, 93: 658–662.
- Kruijt, J.P., 1964. Ontogeny of social behaviour in Burmese Red Junglefowl, (*Gallus gallus spadiceus*). *Behaviour Suppl.*, 12: 201 pp.
- Levine, R.L., Hunter, J.E. and Borchelt, P.L., 1974. Dust bathing as a regulatory mechanism. *Bull. Math. Biol.*, 36: 545–553.
- Martin, G., 1975. Über Verhaltensstörungen von Legehennen im Käfig. *Angew. Ornithol.*, 4: 145–176.
- Nice, M.M., 1962. Development of behaviour in precocial birds. *Trans. Linn. Soc. N.Y.*, 8: 1–211.
- Penquite, R., Craft, W.A. and Thomson, R.B., 1929. Variation in activity and production of spermatozoa by White Leghorn males. *Poult. Sci.*, 9: 247–256.
- Siegel, S., 1956. *Nonparametric Statistics for the Behavioral Sciences*. McGraw-Hill Kogakusha, Tokyo, 312 pp.
- Siegel, P.B. and Guhl, A.M., 1956. The measurement of some diurnal rhythms in activity of White Leghorn hens. *Poult. Sci.*, 35: 1340–1345.
- Simmons, K.E.L., 1964. Feather maintenance. In: A.L. Thompson (Editor), *A New Dictionary of Birds*. McGraw-Hill, New York, pp. 278–286.
- Statkiewicz, W.R. and Schein, M.W., 1980. Variability and periodicity of dust-bathing behaviour in Japanese quail (*Coturnix coturnix japonica*). *Anim. Behav.*, 28: 462–467.
- Vestergaard, K., 1981. Behavioural and physiological studies of hens on wire floors and in deep litter pens. In: D.W. Fölsch (Editor), *Das Verhalten von Legehennen. The Behaviour of Fowl*. Birkhäuser, Basel, Boston, Stuttgart, pp. 115–132.
- Wennrich, G., 1977. Zum Nachweis eines "Triebstaus" bei Haushennen (*Gallus gallus* var. *domesticus*). *Dtsch. Tierärztl. Wochenschr.*, 84: 310–316.
- Wood-Gush, D.G.M., 1959. Time-lapse photography: a technique for studying diurnal rhythms. *Physiol. Zoöl.*, 32: 272–283.